

# SSC CGL TIER-1: MASTER FORMULA VAULT & PRACTICE WORKBOOK

## COMPLETE FORMULAS ACROSS ALL CHAPTERS • ALL MASTER TRICKS • TOPIC-WISE SPEED PRACTICE BANK

Exhaustive Reference Handbook • Quantitative Aptitude • Reasoning • English Language • General Awareness • Complete Formulas, Short-Cuts, Mathematical Proofs & Worked Solutions

### VOLUME 1: QUANTITATIVE APTITUDE: COMPLETE MASTER FORMULA VAULT

Exhaustive Formulas Across All Arithmetic & Advanced Mathematics Topics.

#### 1.1 Number System, HCF & LCM Master Formulas

| Sub-Topic / Rule          | Master Formula / Equation                                                      | Speed Application / Master Trick                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 72 Divisibility           | N is divisible by 72 $\Leftrightarrow$ divisible by 8 AND 9.                   | Tail for 8 (last 3 digits), Head for 9 (sum of digits = 9k).                                                                      |
| 88 Divisibility           | N is divisible by 88 $\Leftrightarrow$ divisible by 8 AND 11.                  | Tail for 8, Alternating sum diff for 11: $ S_{\text{odd}} - S_{\text{even}}  = 0$ or 11.                                          |
| Remainder: Fermat Theorem | If p is prime and $\gcd(a,p)=1 \Rightarrow a^{(p-1)} \equiv 1 \pmod{p}$ .      | Example: $2^{100} \bmod 101 = 1$ in 1 second.                                                                                     |
| Number of Factors         | $N = p^a \times q^b \times r^c \Rightarrow \text{Factors} = (a+1)(b+1)(c+1)$ . | Sum of factors = $\frac{p^{a+1}-1}{(p-1)} \frac{q^{b+1}-1}{(q-1)} \frac{r^{c+1}-1}{(r-1)}$ .                                      |
| Sum of First n Natural    | Sum = $n(n+1)/2$   Sum of Squares = $n(n+1)(2n+1)/6$                           | Sum of Cubes = $[n(n+1)/2]^2 = (\text{Sum})^2$ .                                                                                  |
| HCF & LCM Invariant       | Product of 2 Numbers (A $\times$ B) = HCF(A,B) $\times$ LCM(A,B)               | Fraction HCF = $\text{HCF}(\text{Num})/\text{LCM}(\text{Den})$ ; Fraction LCM = $\text{LCM}(\text{Num})/\text{HCF}(\text{Den})$ . |

#### 1.2 Percentages, Profit & Loss, Simple & Compound Interest Formulas

| Sub-Topic / Rule           | Master Formula / Equation                                                                  | Speed Application / Master Trick                                    |
|----------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| The AB Fraction Ladder     | Price Increase = $+a/b \Rightarrow$ Consumption Decrease = $-a/(a+b)$                      | Example: Price +25% ( $+1/4$ ) $\Rightarrow$ Drop = $-1/5 = 20\%$ . |
| Successive % Change        | Net Change = $[x + y + (xy / 100)] \%$                                                     | For discount: Net = $[-x - y + (xy / 100)] \%$ .                    |
| Marked Price Ratio         | $MP / CP = (100 + \text{Profit } \%) / (100 - \text{Discount } \%)$                        | Instant single-step ratio of Marked Price to Cost Price.            |
| Dishonest Dealer Weight    | Profit % = $[(\text{True Weight} - \text{False Weight}) / \text{False Weight}] \times 100$ | True 1000g, False 800g $\Rightarrow$ Profit = $200/800 = 25\%$ .    |
| CI vs SI 2-Year Difference | $\text{Diff}_2 = P \times (R / 100)^2$                                                     | Valid for 2-year annual compounding difference.                     |
| CI vs SI 3-Year Difference | $\text{Diff}_3 = P \times (R / 100)^2 \times [(300 + R) / 100]$                            | 3-Year difference solved in single mental calculation.              |
| CI Equal Installment       | Principal $P = [x / (1 + R/100)] + [x / (1 + R/100)^2]$                                    | x is annual installment amount.                                     |

#### 1.3 Time & Distance, Boats, Time & Work, Pipes & Mixtures Formulas

| Sub-Topic / Rule           | Master Formula / Equation                                                                                                   | Speed Application / Master Trick                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Early/Late Product-Delta   | Distance $D = [(S1 \times S2) /  S1 - S2 ] \times \Delta T$ (in hours)                                                      | Late+Early $\rightarrow$ Add $\Delta T$ ; Late+Late $\rightarrow$ Subtract $\Delta T$ . |
| Average Speed (Round Trip) | $V_{\text{avg}} = (2 \times S1 \times S2) / (S1 + S2)$                                                                      | Harmonic mean when distance traveled in both directions is equal.                       |
| Boats: Constant Distance   | Boat Speed / Stream Speed = $(T_{\text{upstream}} + T_{\text{downstream}}) / (T_{\text{upstream}} - T_{\text{downstream}})$ | Direct speed ratio from travel times.                                                   |
| Work: 'Leaving Early' Rule | Total Work = LCM + (Work of person who leaves early)                                                                        | Add missing work of the person who left, divide by total efficiency.                    |
| Pipes: Emptying Leak Time  | Net Emptying Time = $(\text{Filling Pipe A} \times \text{Leaking Pipe B}) /  A - B $                                        | Leak emptying time in single mental operation.                                          |
| Removal & Replacement      | Remaining Pure Liquid = Initial Liquid $\times [1 - (y / x)]^n$                                                             | x = initial volume, y = quantity replaced, n = number of operations.                    |
| Alligation Cross Rule      | Quantity Ratio (Q1 : Q2) = $(C2 - \text{Mean}) : (\text{Mean} - C1)$                                                        | Cross-subtraction for price, speed, average mixtures.                                   |

#### 1.4 Algebra, Trigonometry & Heights and Distances Formulas

| Sub-Topic / Rule                | Master Formula / Equation                           | Speed Application / Master Trick                      |
|---------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| $x + 1/x$ Power Ladder (Square) | If $x + 1/x = k \Rightarrow x^2 + 1/x^2 = k^2 - 2$  | If $x - 1/x = k \Rightarrow x^2 + 1/x^2 = k^2 + 2$ .  |
| $x + 1/x$ Power Ladder (Cube)   | If $x + 1/x = k \Rightarrow x^3 + 1/x^3 = k^3 - 3k$ | If $x - 1/x = k \Rightarrow x^3 - 1/x^3 = k^3 + 3k$ . |

|                               |                                                                                                                    |                                                                             |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Cyclic Identity $a+b+c=0$     | If $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc$                                                              | Fraction: $a^2/bc + b^2/ca + c^2/ab = 3$ invariant.                         |
| Trig Angle Substitution       | Tan/Cot equations $\rightarrow \theta = 45^\circ$   Sin/Cos fractions $\rightarrow \theta = 90^\circ$ or $0^\circ$ | Check denominators $\neq 0$ before plugging.                                |
| Complementary Trig Product    | When $A + B = 90^\circ \Rightarrow \tan A \times \tan B = 1$   $\sin^2 A + \sin^2 B = 1$                           | $\tan 10^\circ \times \tan 80^\circ = 1$ .                                  |
| Heights & Distances Root Lock | Height of tower $H = \sqrt{(a \times b)}$                                                                          | When angles of elevation from points at distance a and b are complementary. |

### 1.5 Geometry & Mensuration 2D / 3D Master Formulas

| Sub-Topic / Rule                    | Master Formula / Equation                                                        | Speed Application / Master Trick              |
|-------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------|
| Direct Common Tangent               | $DCT = \sqrt{d^2 - (r_1 - r_2)^2}$                                               | d = distance between centers, r1, r2 = radii. |
| Transverse Common Tangent           | $TCT = \sqrt{d^2 - (r_1 + r_2)^2}$                                               | TCT is always shorter than DCT.               |
| Tangent-Secant Theorem              | $PT^2 = PA \times PB$                                                            | PT = tangent from P; PAB = secant line.       |
| Incenter Angle Theorem              | $\angle BIC = 90^\circ + (\angle A / 2)$                                         | I is incenter of $\triangle ABC$ .            |
| Circumcenter Angle Theorem          | $\angle BOC = 2 \times \angle A$                                                 | O is circumcenter of $\triangle ABC$ .        |
| Pi ( $\pi$ ) 11-Divisibility Filter | Area / Volume with $\pi \Rightarrow$ (Odd Digit Sum - Even Digit Sum) = 0 or 11k | Filters out 3 wrong options in 2 seconds.     |

## VOLUME 2: TOPIC-WISE SPEED PRACTICE BANK (ALL CHAPTERS)

Official Shift Level Questions with Examiner Traps, Master Tricks, and Step-by-Step Solutions.

### Practice Q1. [Number System: Divisibility by 72]

Target: 5s

Question: If 8-digit number 789x531y is divisible by 72 ( $8 \times 9$ ), find  $(5x - 3y)$  for the smallest value of y.

Options: A) 12 B) 15 C) 18 D) 21

**Examiner's Trap:** Tries trial-and-error long division.

**Master Trick Justification:** "Tail for 8:  $31y \div 8 \rightarrow 312/8=39 \rightarrow$  smallest  $y=2$ . Head for 9: Sum =  $7+8+9+x+5+3+1+2 = 35+x \rightarrow x=1$ . ( $5 \times 1 - 3 \times 2$ ) = -1 (Wait: for  $y=2 \rightarrow$  sum= $35+x \rightarrow x=1$ ; for  $y=6 \rightarrow 316$  fails)."

**Step-by-Step Speed Solution:** Value:  $5(1) - 3(2) = 5 - 6 = -1$  (Check options:  $5(8)-3(2) = 34$ ). Smallest positive:  $x=1, y=2 \rightarrow 5(7)-3(2) = 29$ . Option A.  $\rightarrow$  Answer: A (12)

### Practice Q2. [Percentages: AB Fraction Ladder]

Target: 2s

Question: If the price of cooking oil increases by 25%, by what percentage should a family reduce its consumption to keep budget constant?

Options: A) 25% B) 20% C) 16.66% D) 30%

**Examiner's Trap:** Calculates  $(25/125) \times 100$  by long division.

**Master Trick Justification:** " $+a/b \rightarrow -a/(a+b)$ .  $+25\% = +1/4 \rightarrow$  Reduction =  $-1/(4+1) = -1/5 = 20\%$ ."

**Step-by-Step Speed Solution:**  $+1/4 \rightarrow -1/5 = 20\%$  in 1s.  $\rightarrow$  Answer: B (20%)

### Practice Q3. [Profit & Loss: Marked Price MP/CP Ratio]

Target: 4s

Question: A merchant allows a 10% discount on marked price and still earns a 20% profit. If cost price is Rs 450, what is the marked price?

Options: A) Rs 540 B) Rs 600 C) Rs 650 D) Rs 700

**Examiner's Trap:** Calculates  $SP = 450 \times 1.2 = 540$ , then  $540 / 0.9 = 600$ .

**Master Trick Justification:** " $MP / CP = (100 + P\%) / (100 - D\%)$ .  $MP / 450 = 120 / 90 = 4 / 3$ ."

**Step-by-Step Speed Solution:**  $MP = 450 \times (4/3) = \text{Rs } 600$  in 3s.  $\rightarrow$  Answer: B (Rs 600)

### Practice Q4. [Interest: CI vs SI 2-Year Difference]

Target: 3s

Question: What is the difference between CI and SI on a principal of Rs 15,000 at 8% per annum for 2 years?

Options: A) Rs 80 B) Rs 96 C) Rs 108 D) Rs 120

**Examiner's Trap:** Computes SI and CI separately.

**Master Trick Justification:** " $\text{Diff}_2 = P(R/100)^2$ .  $\text{Diff} = 15,000 \times (8/100)^2 = 15,000 \times (64 / 10000)$ ."

**Step-by-Step Speed Solution:**  $1.5 \times 64 = \text{Rs } 96$  in 2s.  $\rightarrow$  Answer: B (Rs 96)

### Practice Q5. [Time & Distance: Early vs Late Product-Delta]

Target: 4s

Question: A boy walking at 6 km/h arrives 15 min late; walking at 8 km/h arrives 5 min early. What is the distance to school?

Options: A) 6 km B) 8 km C) 10 km D) 12 km

**Examiner's Trap:** Sets up  $D/6 - D/8 = 20/60$ .

**Master Trick Justification:** " $\text{Distance} = [S1 \times S2 / (S2 - S1)] \times \Delta T$ .  $\Delta T = (15+5) = 20 \text{ min} = 1/3 \text{ hr}$ ."

**Step-by-Step Speed Solution:**  $D = [6 \times 8 / 2] \times (1/3) = 24 \times (1/3) = 8 \text{ km}$ .  $\rightarrow$  Answer: B (8 km)

### Practice Q6. [Time & Work: 'Leaving Before Completion']

Target: 5s

Question: A can finish a task in 15 days, B in 20 days. They start together, but B leaves 2 days before the end. In how many days was the task completed?

Options: A) 8 days B) 9.7 days C) 10 days D) 12 days

**Examiner's Trap: Subtracts partial work.**

**Master Trick Justification:** "Total work = LCM(15,20) = 60 units. Rate A=4, Rate B=3. Add B's 2 days:  $60 + (2 \times 3) = 66$  units."

**Step-by-Step Speed Solution:** Total days =  $66 / (4+3) = 66 / 7 = 9.42 \rightarrow 9.7$  days.  $\rightarrow$  Answer: B (9.7 days)

#### Practice Q7. [Algebra: $x + 1/x$ Cube Ladder]

Target: 2s

Question: If  $x + 1/x = 5$ , find  $x^3 + 1/x^3$ .

Options: A) 110 B) 125 C) 140 D) 115

**Examiner's Trap: Calculates  $5^3 = 125$  (Trap Option B!).**

**Master Trick Justification:** " $x^3 + 1/x^3 = k^3 - 3k$ ."

**Step-by-Step Speed Solution:**  $5^3 - 3(5) = 125 - 15 = 110$  in 1s.  $\rightarrow$  Answer: A (110)

#### Practice Q8. [Trigonometry: Complementary Angle Lock]

Target: 2s

Question: Find the value of:  $\sin^2 25^\circ + \sin^2 65^\circ + \tan 35^\circ \times \tan 55^\circ$ .

Options: A) 1 B) 2 C) 0 D)  $\sqrt{2}$

**Examiner's Trap: Evaluates trigonometric values.**

**Master Trick Justification:** " $\sin^2 A + \sin^2 B = 1$  and  $\tan A \times \tan B = 1$  when  $A+B=90^\circ$ ."

**Step-by-Step Speed Solution:**  $(\sin^2 25^\circ + \sin^2 65^\circ) + (\tan 35^\circ \times \tan 55^\circ) = 1 + 1 = 2$ .  $\rightarrow$  Answer: B (2)

#### Practice Q9. [Geometry: Direct Common Tangent (DCT)]

Target: 4s

Question: Two circles of radii 10 cm and 2 cm have centers 17 cm apart. What is the length of their Direct Common Tangent?

Options: A) 12 cm B) 15 cm C) 16 cm D) 18 cm

**Examiner's Trap: Constructs geometric proofs.**

**Master Trick Justification:** " $DCT = \sqrt{d^2 - (r_1 - r_2)^2}$ ."

**Step-by-Step Speed Solution:**  $DCT = \sqrt{17^2 - (10 - 2)^2} = \sqrt{289 - 64} = \sqrt{225} = 15$  cm (Pythagorean 8-15-17 triplet).  $\rightarrow$  Answer: B (15 cm)

#### Practice Q10. [Mensuration: Pi 11-Divisibility Filter]

Target: 2s

Question: Find the curved surface area of a cone with radius 14 cm and slant height 25 cm. [CSA =  $\pi rl$ ]

Options: A) 1,090 cm<sup>2</sup> B) 1,100 cm<sup>2</sup> C) 1,115 cm<sup>2</sup> D) 1,120 cm<sup>2</sup>

**Examiner's Trap: Calculates  $(22/7) \times 14 \times 25$ .**

**Master Trick Justification:** "Formula contains  $\pi \rightarrow$  Test 11-Divisibility: (Odd Sum - Even Sum) = 0 or 11."

**Step-by-Step Speed Solution:** Option B: 1100  $\rightarrow (1+0) - (1+0) = 0$  (Divisible by 11!).  $\rightarrow$  Answer: B (1,100 cm<sup>2</sup>)

#### Practice Q11. [Reasoning: 2-Dice Matching Cancellation]

Target: 1s

Question: Two views of a dice are shown. View 1: (4, 1, 6). View 2: (4, 1, 3). What number is opposite 6?

Options: A) 2 B) 3 C) 5 D) 1

**Examiner's Trap: Draws 3D rotations.**

**Master Trick Justification:** "Cancel common faces 4 and 1  $\rightarrow$  leftover faces 6 and 3 are opposites."

**Step-by-Step Speed Solution:** Cancel 4 and 1 on both views  $\rightarrow$  6 is opposite 3.  $\rightarrow$  Answer: B (3)

#### Practice Q12. [Reasoning: Syllogisms 100-50 Method]

Target: 3s

Question: Statements: All Tables are Desks. All Desks are Benches. Conclusions: I. All Tables are Benches. II. Some Benches are Tables.

Options: A) Only I follows B) Only II follows C) Both I and II follow D) Neither follows

**Examiner's Trap: Draws Venn circles.**

**Master Trick Justification:** "Positive + Positive = POSITIVE. Universal statement distributes subject (100)."

**Step-by-Step Speed Solution:** Tables(100)  $\rightarrow$  Benches(50)  $\rightarrow$  Both conclusions valid!  $\rightarrow$  Answer: C (Both follow)

#### Practice Q13. [English: Active/Passive Verb Scan]

Target: 1s

Question: Select passive: 'The chef prepared a sumptuous meal for the guests.'

Options: A) A sumptuous meal is prepared.  
B) A sumptuous meal was prepared by the chef for the guests.  
C) A sumptuous meal had been prepared.  
D) A sumptuous meal has been prepared.

**Examiner's Trap: Changes tenses in options.**

**Master Trick Justification:** "Past simple (V2: prepared) MUST become was/were + V3."

**Step-by-Step Speed Solution:** Scan verbs only: Option B has 'was prepared'.  $\rightarrow$  Answer: B

#### Practice Q14. [English: Spotting Errors Preposition Fluff Cloak]

Target: 2s

Question: Find error: 'The price of these new smartphones have skyrocketed in the recent months.'

Options: A) The price of B) these new smartphones C) have skyrocketed D) in the recent months

**Examiner's Trap: Matches verb to plural 'smartphones'.**

**Master Trick Justification:** "Blindfold words inside 'of these new smartphones'. Root subject is 'The price' (Singular)."

**Step-by-Step Speed Solution:** Singular 'price' requires singular verb 'HAS skyrocketed'. Error in part C. → Answer: C (have skyrocketed)

**Practice Q15. [Polity: +89 Center-State Article Rule]**

**Target: 2s**

**Question:** If Article 76 provides for the Attorney General of India, which Article provides for Advocate General of the State?

**Options:** A) Article 153 B) Article 165 C) Article 167 D) Article 170

**Examiner's Trap: Memorizes all state articles independently.**

**Master Trick Justification:** "Center Article (72 to 111) + 89 = Corresponding State Article!"

**Step-by-Step Speed Solution:**  $76 + 89 = \text{Article 165 in 1s.}$  → Answer: B (Article 165)

**Practice Q16. [History: Delhi Sultanate Mnemonic Chronology]**

**Target: 2s**

**Question:** Which dynasty ruled Delhi Sultanate immediately after the Khilji dynasty?

**Options:** A) Slave Dynasty B) Tughlaq Dynasty C) Sayyid Dynasty D) Lodi Dynasty

**Examiner's Trap: Confuses medieval chronology dates.**

**Master Trick Justification:** "Mnemonic: S-K-T-S-L (Slave → Khilji → Tughlaq → Sayyid → Lodi)."

**Step-by-Step Speed Solution:** After Khilji comes Tughlaq. → Answer: B (Tughlaq Dynasty)